RAJALAKSHMI ENGINEERING COLLEGE

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M2

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| **Started on** | Thursday, 27 August 2026, 11:02 AM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 11:45 AM |
| **Time taken** | 42 mins 37 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

A person needs to eat burgers. Each burger contains a count of calorie. After eating the burger, the person needs to run a distance to burn out his calories.

If he has eaten $i$ burgers with c calories each, then he has to run at least $3^i * c$ kilometers to burn out the calories. For example, if he ate 3

burgers with the count of calorie in the order: [1, 3, 2], the kilometers he needs to run are $(3^0 * 1) + (3^1 * 3) + (3^2 * 2)$= $1 + 9 + 18 = 28$.

But this is not the minimum, so need to try out other orders of consumption and choose the minimum value. Determine the minimum distance

he needs to run. Note: He can eat burger in any order and use an efficient sorting algorithm.Apply greedy approach to solve the problem.

**Input Format**

First Line contains the number of burgers

Second line contains calories of each burger which is n space-separate integers

**Output Format**

Print: Minimum number of kilometers needed to run to burn out the calories

**Sample Input**

```
3
5 10 7
```

**Sample Output**

```
76
```

**For example:**

| Test | Input | Result |
|---|---|---|
| Test Case 1 | 3<br>1 3 2 | 18 |

**Answer:** (penalty regime: 0 %)

```
#include<stdio.h>
#include<math.h>
int main(){
    int n;
    scanf("%d",&n);
    int arr[n];
    for(int i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    int temp;
    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(arr[i]<arr[j]){
                temp=arr[i];
                arr[i]=arr[j];
                arr[j]=temp;
            }
        }
    }
    int sum=0;
```

```
    for(int i 0;i<n;i++){
        sum=sum+pow(n,i)*arr[i];
    }
    printf("%d",sum);
}
```

| | Test | Input | Expected | Got | |
|---|---|---|---|---|---|
| | Test Case 1 | 3<br>1 3 2 | 18 | 18 | |
| | Test Case 2 | 4<br>7 4 9 6 | 389 | 389 | |
| | Test Case 3 | 3<br>5 10 7 | 76 | 76 | |

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

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